

Bayesian Data Analysis

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Topic 2:

From Prior Information to Posterior Inference

Bayesian learning

- Bayesians quantify uncertainty about fixed but unknown **parameters** by treating them as **random variables**.
- This requires us to set a **prior distribution** $p(\theta)$ to summarize uncertainty **before observing the data**.
- The **distribution** of the **observed data given the model parameters** is the **likelihood function** $p(y|\theta)$.
 - The likelihood function is the **most important** piece of a Bayesian analysis because it **links the data and parameters**.

Bayesian learning

- The **posterior distribution** $p(\theta|y)$ summarizes uncertainty about the parameters **given the prior and data**
- The **reduction in uncertainty** from prior to posterior represents **Bayesian learning**.
- **Bayes' Theorem (Bayes' Rule)** converts the **likelihood and prior to the posterior**:

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)}$$

where $p(y)$ is **the marginal distribution** of the data and can usually be **ignored**.

How to select a prior?

- There is **no “true” or “correct” prior**.
 - In some cases expert opinion or similar studies can be used to specify an **informative prior**.
 - If prior information is unavailable, then the prior should be **uninformative**.
- The prior is best viewed as an **initial value** to a statistical procedure.
- However, there is inherent **subjectivity** to selecting the prior
 - That is, different analysts may pick **different priors** and thus have **different results**

How to select a likelihood?

- The likelihood is the **same as** in a **frequentist analysis**.
- Main consideration in selecting a likelihood is the **type of data** and the **support**.
- For example, in a linear regression analysis we may say:

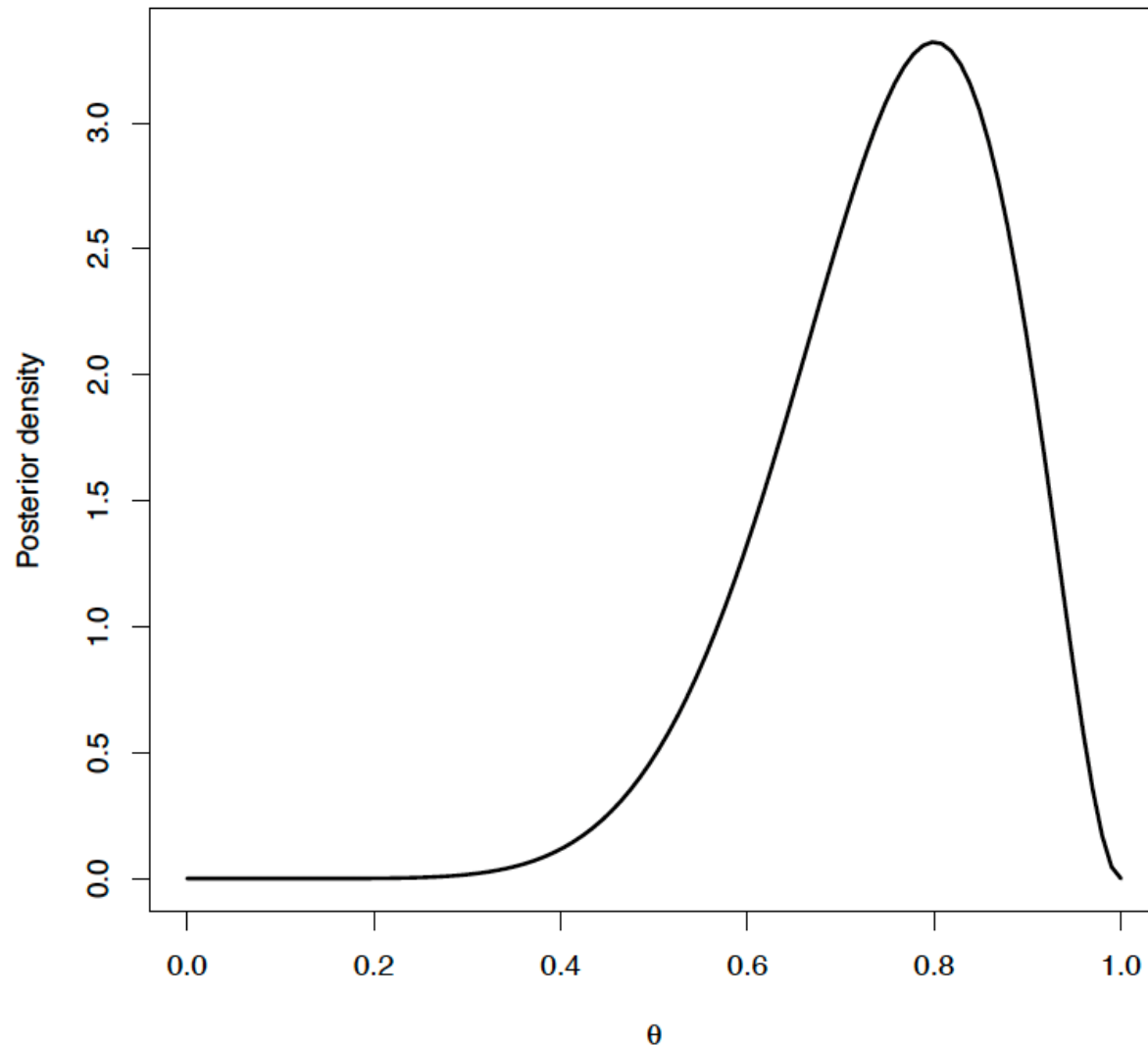
$$Y_i | \boldsymbol{\beta}, \sigma^2 \sim \text{Normal} \left(\sum_{j=1}^p X_{ij} \beta_j, \sigma^2 \right)$$

Summarizing a posterior

Summarizing a univariate posterior

- After selecting the likelihood and prior, all that remains is to **summarize the posterior**.
- Say there is a single parameter, θ
- For example, we say the model is
Likelihood: $Y|\theta \sim \text{Binomial}(n, \theta)$
Prior: $\theta \sim \text{Uniform}(0,1)$
- We saw that the posterior is then
 $\theta|Y \sim \text{Beta}(Y + a, n - Y + b)$
- The posterior is a distribution that can be plotted as on the next slide.

Summarizing a univariate posterior



In this beta/binomial example $Y = 8$, $n = 10$, and $a = b = 1$.

However, to be more concise, we typically use a few **numerical summaries** of the distribution.

The posterior can be summarized like any other distribution, e.g. mean, variance, skewness, etc.

Point estimators

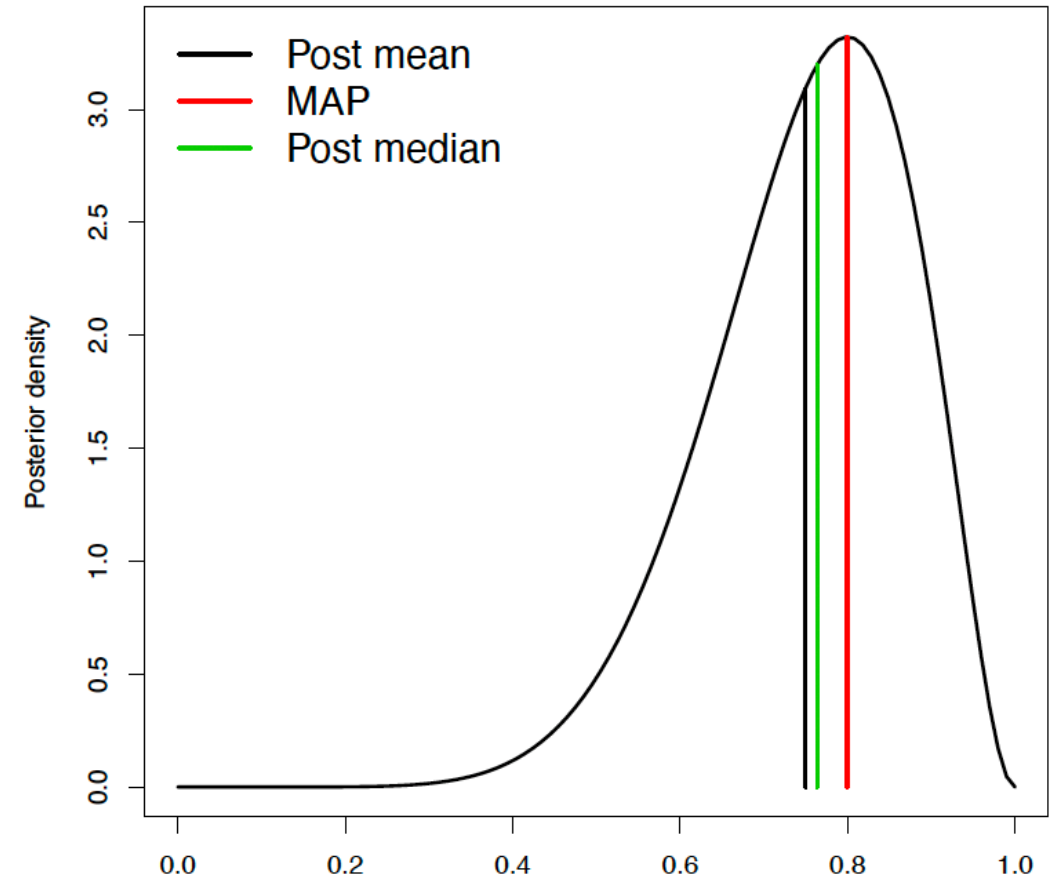
- A **point estimator** is a one number summary used to estimate the unknown parameter.
- For example, we might use the **posterior mean (or median)** as the “**best guess**” of θ .
- The posterior mean is

$$\hat{\theta} = E(\theta|Y) = \int \theta p(\theta|Y) d\theta$$

- For the **beta/binomial** example $\hat{\theta} = \frac{Y+a}{n+a+b}$ Can be from cheatshit of a distribution
- This is an alternative to the sample proportion $\hat{\theta} = \frac{Y}{n}$

MAP estimator

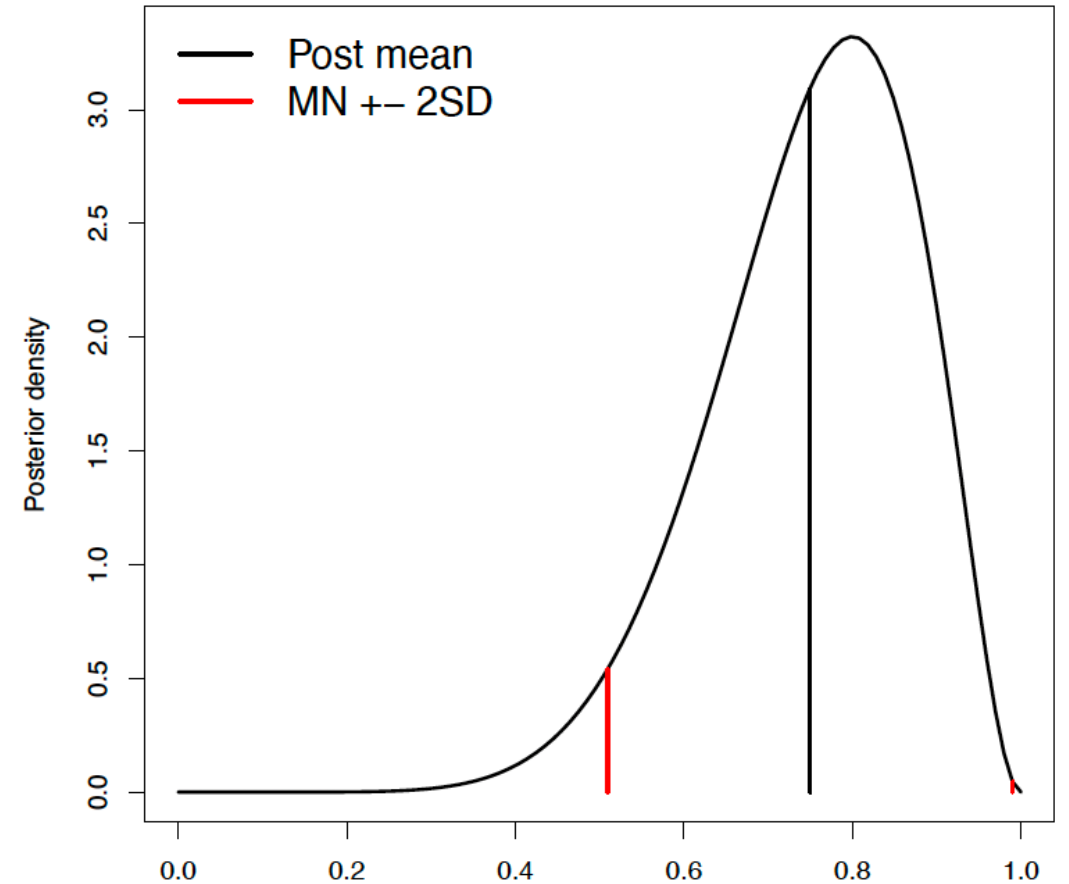
- The **posterior mode** is called the **maximum a posteriori (MAP) estimator**, which is defined as the value that maximizes the posterior.
- The MAP estimator is
$$\hat{\theta} = \arg \max_{\theta} \log[p(\theta|Y)]$$
Theta that maximized posterior density
$$= \arg \max_{\theta} \log[p(Y|\theta)] + \log[p(\theta)]$$
- If the prior is uniform (flat) the MAP is the MLE (Maximum Likelihood Estimator).



In this beta/binomial ^{θ} example $Y = 8$, $n = 10$, and $a = b = 1$.

Posterior standard deviation

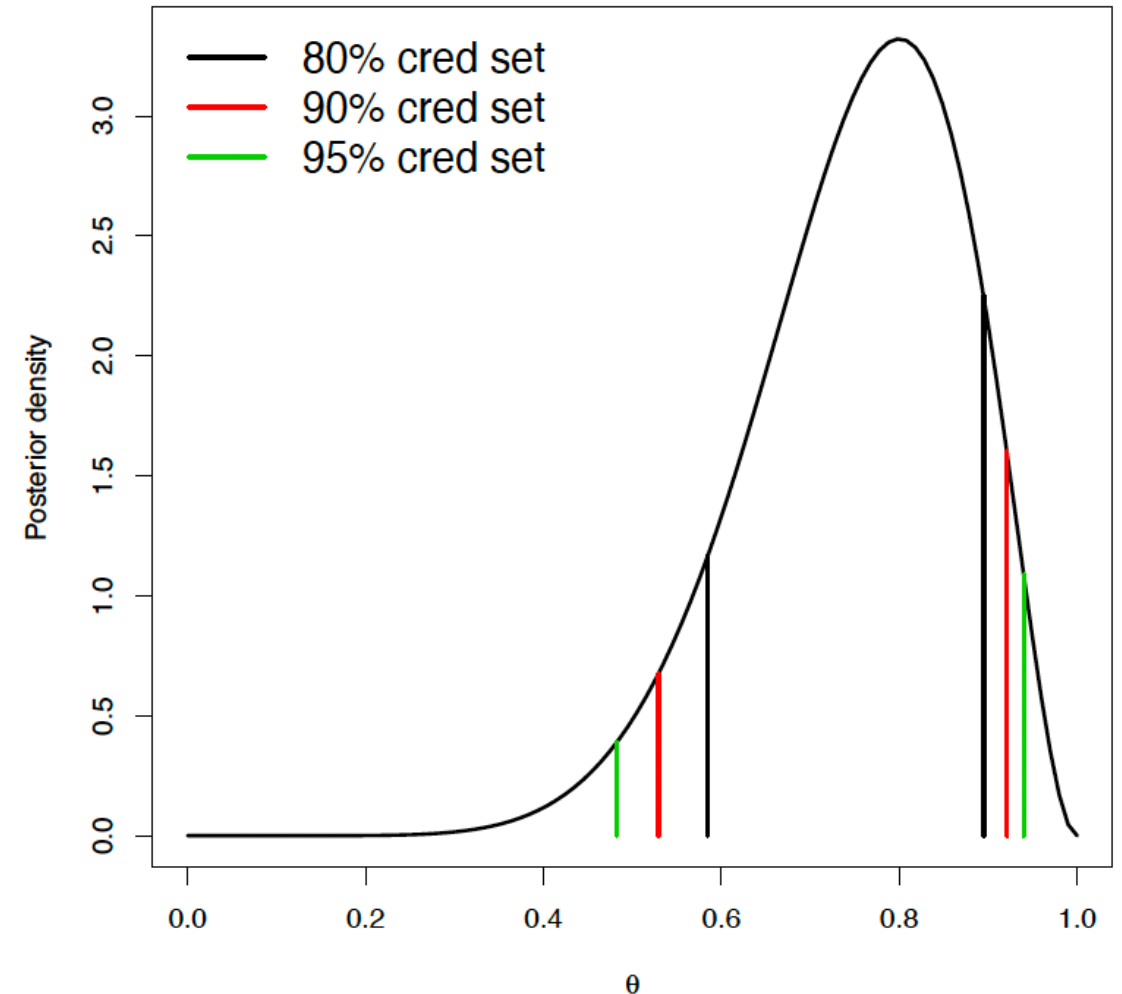
- The **posterior standard deviation (SD)** is one **measure of uncertainty**.
- If the posterior is approximately normal, then the **mean plus/minus two SD** units captures **95% of the posterior probability**.
- The posterior SD is analogous but fundamentally different than the frequentist **standard error (SE)**.
- The SE is the standard deviation of $\hat{\theta}$'s sampling distribution.



In this beta/binomial^θ example $Y = 8$, $n = 10$, and $a = b = 1$.

Credible intervals

- In addition to standard error, uncertainty can be quantified using a **credible interval**.
- The interval (l, u) is a $100(1 - \alpha)\%$ posterior credible interval/set if
$$Prob(l < \theta < u | Y) = 1 - \alpha$$
- Interpretation of a 95% credible interval: **“given the data and prior, I am 95% certain that θ is between l and u ”**
- The **equal-tailed interval** is $(q_{\alpha/2}, q_{1-\alpha/2})$
- This is analogous but different than a **confidence interval**.



In this beta/binomial example $Y = 8$, $n = 10$, and $a = b = 1$.

Posterior probability

- **Hypothesis tests** are conducted by simply computing the **posterior probability** of each hypothesis.
- Say the null hypothesis is $H_0: \theta \leq 0.5$ and the alternative is $H_1: \theta > 0.5$.
- The **posterior probability of the null hypothesis** is

$$Prob(\theta < 0.5|Y) = \int_0^{0.5} p(\theta|Y) d\theta$$

- We **reject** the null if its probability is small.
- In a Bayesian analysis we can say “**Given the data and prior, the probability that the null hypothesis is true is 0.03**”.
- This is analogous to but different than the **p-value**.

Monte Carlo approximations

- **Monte Carlo (MC) sampling** is a useful tool for summarizing a posterior.
- For univariate cases, it is not particularly useful but in harder problems is the best approach available.
- In MC sampling, we **draw S samples** from the posterior,
$$\theta^{(1)}, \dots, \theta^{(S)} \sim p(\theta|Y)$$
and use these samples to **approximate the posterior**.
- For example, the posterior mean and variance are approximated by the sample mean and variance of the $\theta^{(s)}$.

Monte Carlo approximations

- MC sampling facilitates studying **transformations of parameters**.
- For example, the **odds** corresponding to θ are $\gamma = \theta / (1 - \theta)$
- How to approximate the posterior mean and variance γ ?
- We simply transform each draw to the odds?

$$\gamma^{(1)} = \frac{\theta^{(1)}}{1 - \theta^{(1)}}, \dots, \gamma^{(S)} = \frac{\theta^{(S)}}{1 - \theta^{(S)}}$$

and use these draws to approximate γ 's posterior.

Summarizing multivariate posteriors

- Say $\boldsymbol{\theta} = (\theta_1, \dots, \theta_p)$
- Ideally, we reduce to the **univariate marginal posteriors**

$$p(\theta_1|Y) = \int \dots \int p(\theta_1, \dots, \theta_p|Y) d\theta_2, \dots, d\theta_p$$

- Then, the same ideas we used for univariate posteriors apply
- However, computing these integrals is often **challenging**.
- **MC sampling** will be a key tool (we will discuss this further in later topics)

Conjugate priors

Selecting priors

- Selecting the prior is one of the **most important steps** in a Bayesian analysis
- There is **no** “right” way to select a prior
- The choices often depend on the **objective of the study** and the **nature of the data**
 - Conjugate versus non-conjugate
 - Informative versus uninformative
 - Proper versus improper
 - Subjective versus objective

Conjugate priors

- A prior is **conjugate** if the posterior is a member of the **same parametric family**.
- We have seen that if the **response is binomial** and we use a **beta prior**, the **posterior is also a beta**.
- This requires a pairing of the likelihood and prior.
- There is a long list of conjugate priors:
https://en.wikipedia.org/wiki/Conjugate_prior
- The advantage of a conjugate prior is that the **posterior is available in closed form**.

Conjugate priors

- Here is an example of a **non-conjugate prior**
- Say $Y \sim \text{Poisson}(\lambda)$ and $\lambda \sim \text{Beta}(a, b)$
- The **posterior** is

$$f(\lambda|Y) \propto \left\{ \exp(-\lambda) \lambda^Y \right\} \left\{ \lambda^{a-1} (1 - \lambda)^{b-1} \right\}$$

- This is **not** a beta PDF, so the prior is not conjugate.
- In fact, this is not a member of any known family of distributions.

Conjugate priors: beta-binomial model

Beta-binomial model

- We use beta-binomial model to **estimate a proportion**, for example:
 - What is the success probability of a new cancer treatment?
 - What proportion of voters support my candidate?
 - What proportion of the population has a rare gene?
- Let $\theta \in [0,1]$ be the proportion we are trying to estimate (e.g. the success probability)
- We conduct n independent trials, each with success probability θ , and observe $Y \in \{0, \dots, n\}$
- We would like to obtain the posterior of θ , a 95% interval, and a test that θ equals some predetermined value θ_0 .

Frequentist analysis

- The maximum likelihood estimate (for proportion) is the sample proportion $\hat{\theta} = \frac{Y}{n}$.

- For large Y and $n - Y$, the sampling distribution of $\hat{\theta}$ is approximately

$$\hat{\theta} \sim \text{Normal} \left(\theta, \frac{\theta(1 - \theta)}{n} \right)$$

- The standard error (standard deviation of the sampling distribution) is approximated as

$$SE(\hat{\theta}) \approx \sqrt{\frac{\hat{\theta}(1 - \hat{\theta})}{n}}$$

- A 95% CI is then $\hat{\theta} \pm 2SE(\hat{\theta})$

Bayesian analysis – likelihood

- Since Y is the number of successes in n independent trials, each with success probability θ , its distribution is

$$Y|\theta \sim \text{Binomial}(n, \theta)$$

- PMF: $P(Y = y|\theta) = \binom{n}{y} \theta^y (1 - \theta)^{n-y}$
- Mean: $E(Y|\theta) = n\theta$
- Variance: $\text{Var}(Y|\theta) = n\theta(1 - \theta)$

Bayesian analysis - prior

- The parameter θ is continuous and between 0 and 1, therefore a natural prior is

$$\theta \sim \text{Beta}(a, b)$$

- PMF: $f(\theta) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1} (1 - \theta)^{b-1}$
- Mean: $E(\theta) = \frac{a}{a+b}$
- Variance: $\text{Var}(\theta) = \frac{ab}{(a+b)^2(a+b+1)}$

Derivation of the posterior

The posterior is $\theta|Y \sim \text{Beta}(a + Y, b + n - Y)$. How to get this?

1. Identify the likelihood and the prior

The likelihood is $p(Y|\theta) = \binom{n}{y} \theta^Y (1 - \theta)^{n-Y}$

The prior is $p(\theta) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1} (1 - \theta)^{b-1}$

2. Use Bayes' Theorem to compute the posterior (try this)

Derivation of the posterior

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2. Use Bayes' Theorem to compute the posterior (try this)

$$\begin{aligned} p(\theta|y) &= \frac{p(y|\theta)p(\theta)}{p(y)} \\ &= \frac{\left[\binom{n}{y} \theta^Y (1 - \theta)^{n-Y} \right] \left[\frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1} (1 - \theta)^{b-1} \right]}{p(y)} \end{aligned}$$

Derivation of the posterior

- Some mathematical tricks give:

$$\begin{aligned} p(\theta|Y) &= \left[\binom{n}{y} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \frac{1}{p(y)} \right] \theta^{Y+a-1} (1-\theta)^{n-Y+b-1} \\ &= C \theta^{A-1} (1-\theta)^{B-1} \end{aligned}$$

where $A = Y + a$, $B = n - Y + b$, and C is the mess.

- The terms that involve θ ,

$$\theta^{A-1} (1-\theta)^{B-1}$$

are the **kernel** of a Beta(A,B) distribution.

- Therefore, $\theta|Y \sim \text{Beta}(Y + a, n - Y + b)$

Simplifying the derivation of the posterior

- In the end, we are always going to look at the **terms that involve θ (the kernel)** and find a **matching distribution**
- Therefore, the mess (C) will always be ignored.
- We simplify the derivation by absorbing **all terms that do not include θ** into a normalizing **constant**.
- For example, instead of

$$p(\theta|Y) = C\theta^{A-1}(1 - \theta)^{B-1}$$

we can write

$$p(\theta|Y) \propto \theta^{A-1}(1 - \theta)^{B-1}$$

Simplifying the derivation of the posterior

- Here is a much simpler derivation

$$\begin{aligned} p(\theta|Y) &\propto p(Y|\theta)p(\theta) \\ &\propto [\theta^Y (1 - \theta)^{n-Y}] [\theta^{a-1} (1 - \theta)^{b-1}] \\ &\propto \theta^{A-1} (1 - \theta)^{B-1} \end{aligned}$$

where $A = Y + a$, $B = n - Y + b$

- Therefore, $\theta|Y \sim \text{Beta}(Y + a, n - Y + b)$
- Note: $p(y)$ was dropped in the first line, and thus is excluded from all these computations.

Selecting the prior

- The posterior is $\theta|Y \sim \text{Beta}(Y + a, n - Y + b)$
- Therefore, a and b can be interpreted as the “**prior number of success and failures**”
- What prior to select if we have **no information** about θ before collecting data?
- What prior to select if **historical data/expert opinion** indicates that θ is likely between 0.6 and 0.8?

Related problem

- The success probability of independent trials is θ .
- Y is the number of event of interests before we observe n other events.

- Then $Y|\theta \sim \text{NegativeBinomial}(n, \theta)$ and

$$\text{Prob}(Y = y|\theta) = \binom{y + n - 1}{y} \theta^n (1 - \theta)^y$$

- Assume the prior $\theta \sim \text{Beta}(a, b)$
- Find the posterior

Related problem

- The **likelihood** is $p(y|\theta) \propto \theta^y (1 - \theta)^n$
- The **prior** is $p(\theta) \propto \theta^{a-1} (1 - \theta)^{b-1}$
- Therefore, the **posterior** is

$$\begin{aligned} p(\theta|y) &\propto [\theta^y (1 - \theta)^n] [\theta^{a-1} (1 - \theta)^{b-1}] \\ &= \theta^{A-1} (1 - \theta)^{B-1} \end{aligned}$$

where $A = y + a$ and $B = n + b$

- This is the **kernel of the beta distribution**, $\theta|y \sim \text{Beta}(A, B)$

Smoking example

- Two smokers have just quit.
- Say subject i has probability θ_i of abstaining each day.
- The number of days until relapse for two patients is 3 and 30 days.
- Can we conclude the patients have different probabilities of relapse?
- What is the probability that their next attempts will exceed 30 days?

Smoking example

- The likelihood is
 $Y_i \sim \text{NegativeBinomial}(1, \theta_i)$
- Assume uniform priors
 $\theta_i \sim \text{Beta}(1, 1)$
- The posteriors are
 $\theta_i | Y_i \sim \text{Beta}(Y_i + 1, 2)$
- Plot the posterior.
- Use Monte Carlo sampling to answer the two questions in previous slide.

